MEP Subject Tracking

Technical Guide

Per-subject home range, speed, seasonal, and occupancy analysis

Generated August 27, 2026

Workflow id: **mep_collared_report**

1. Overview

The **mep_collared_report** workflow fetches GPS relocations and MEP collar events for a named subject group from EarthRanger, derives trajectories, and produces a comprehensive per-subject movement ecology report overlaid on LandDX land-use layers. Google Earth Engine is used to detect NDVI-based seasonal windows that stratify all spatial and temporal analyses.

For each subject the workflow delivers:

- 3 maps — speed map, ETD home range map, and seasonal home range map (with MCP overlay), each overlaid with Community Conservancy, National Reserve, and National Park land-use layers
- 4 time-series plots — Net Square Displacement (NSD), speed, collar event, and MCP asymptote
- 5 single-value dashboard metrics — MCP area, ETD area, distance travelled, max displacement, and night/day ratio
- 3 dashboard table widgets — subject info, subject stats, and land-use occupancy
- 4 CSV tables — subject information, subject stats, occupancy, and seasonal windows
- A Word mapbook — cover page (with organisation logo) + one populated section per subject, merged into a single report

Output summary

Output type	Count	Description
Speed maps	1 per subject	Path layer coloured by 6-bin speed classification
Home range maps	1 per subject	ETD percentile polygons (50–99.9th)
Seasonal home range maps	1 per subject	ETD 99.9th percentile split by wet/dry season + MCP outline
NSD seasonal plots	1 per subject	Net Square Displacement with season bands
Speed seasonal plots	1 per subject	Speed time series with season bands
Collar event plots	1 per subject	Collar check/collaring/source failure events on timeline
MCP asymptote plots	1 per subject	MCP area growth curve
Subject CSV tables	4 per subject	info, stats, occupancy, seasonal windows
Dashboard SV widgets	5 per subject	MCP, ETD, distance, max displacement, night/day ratio
Dashboard table widgets	3 per subject	Subject info, subject stats, subject occupancy
Word mapbook	1 total	Cover page + per-subject report sections merged

2. Dependencies

2.1 Python packages

Package	Version	Channel
ecoscope-platform	>=2.15.0, <2.16.0	ecoscope-workflows
ecoscope-workflows-ext-custom	0.1.0rc14.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ste	0.0.0rc1.*	ecoscope-workflows-custom
ecoscope-workflows-ext-mep	1.0.0.*	ecoscope-workflows-custom
pydeck	0.9.2	conda-forge
opentelemetry-sdk	>=1.20.0, <2.0.0	conda-forge

2.2 Connections

Connection	Task	Purpose
EarthRanger	set_er_connection	Fetch subject groups, GPS relocations, and MEP collar events
Google Earth Engine	set_gee_connection	Compute NDVI-based seasonal windows per subject ETD extent

2.3 LandDx database

The workflow downloads the LandDX GeoPackage automatically via **fetch_and_persist_file** (ecoscope_workflows_ext_ste) from a fixed Dropbox URL (land_dx.gpkg) — this is not user-configurable. The output path is resolved from **ECOSCOPE_WORKFLOWS_RESULTS**. The loaded GeoDataFrame is filtered by **type** into three separate DataFrames, each rendered as its own **create_geojson_layer** with its own legend entry:

LandDx type	Fill/line color (RGBA)	Hex	Opacity	Line width
Community Conservancy	[166, 182, 151, 255]	#a6b697	0.15	1.55 px
National Reserve	[136, 167, 142, 255]	#88a78e	0.15	1.55 px
National Park	[17, 86, 49, 255]	#115631	0.15	1.55 px

Each of the three land-use layers is combined with the per-subject speed, home range, and seasonal home range layers via **combine_deckgl_map_layers** before rendering, so all three maps share the same land-use context.

2.4 Base maps

Base map tiles are configured by the user via **set_base_maps_pydeck**. Any tile layer supported by deck.gl can be specified.

2.5 Grouper

The workflow groups all data by **subject_name**. The rjsf-overrides block restricts the grouper UI to Subject Name only — users cannot change the grouping dimension.

2.6 Subject group

A user-provided string parameter (**Subject Group Name**, default: *Elephants*) is passed to both the subject info fetch and the observations fetch to scope the analysis to a single EarthRanger subject group.

3. Subject Data Pipeline

Subject metadata is fetched and cleaned separately from GPS relocations. Both pipelines share the same subject group name and EarthRanger client.

3.1 Subject metadata

Subjects are fetched by ID rather than by group, using the distinct subject IDs already present in the relocations (see 3.3):

Step	Task	Detail
1	dataframe_column_unique (ext_mep)	Extract unique subject_id values from the mapped relocations
2	get_subjects (ext_mep)	Fetch subjects by ID (include_inactive: true, max_ids_per_request: 50, raise_on_empty: false)
3	normalize_json_column	Flatten the 'additional' JSON column; skip if column absent
4	map_columns	Drop 15 columns (content_type, common_name, image_url, url, tracks_available, last_position_status/date/position, device_status_properties, hex, subject_type, subject_subtype, user, sex, and raw additional__ id fields); rename ~11 columns (id→groupby_col, name→subject_name, hex→hex_color, additional__rgb→rgb, additional__Bio→subject_bio, additional__sex→subject_sex, additional__DOB→date_of_birth, additional__notes→notes, additional__status→status, additional__region→region, additional__country→country, additional__id_photo→photo, additional__distribution→distribution)

3.2 Subject maturity

compute_subject_maturity is called with **months_duration: 6** against the cleaned subject metadata and relocations GeoDataFrame. This flags subjects that have been tracked for at least 6 months, surfaced as the **Mature** field in the subject stats table.

3.3 GPS relocations

Step	Task	Detail
1	get_subjectgroup_observations	Fetch observations for the subject group (filter: clean, time_range, raise_on_empty: false, include_details: false, include_subjectsource_details: false)
2	get_timezone_from_time_range → convert_values_to_timezone	Convert the fixtime column to the analysis timezone

Step	Task	Detail
3	process_relocations	Retain 15 columns including extra__subject_id, extra__id, extra__location, extra__created_at, extra__recorded_at, extra__source, extra__exclusion_flags, extra__subject__updated_at, extra__subject__name, extra__subject__hex, extra__subject__sex, extra__subject__subject_subtype, geometry, groupby_col, fixtime, junk_status. Filter 3 invalid coordinate pairs: (180,90), (0,0), (1,1)
4	map_columns	Rename name→subject_name

3.4 Per-subject outputs

After subjects are split by group (**split_groups** on subject_name):

- **persist_subject_photo** — downloads the subject's profile photo from EarthRanger and saves as PNG (overwrite_existing: true)
- **process_subject_information** — formats subject metadata (maxlen: 1000), persists as CSV, and is transposed into a "Subject Info" table widget for the dashboard

4. Event Data Pipeline

Three MEP-specific event types are fetched from EarthRanger for the analysis time range and used in the collar event plot:

Event type	Description
mep_collar_check	Periodic collar condition check
mep_collaring	Initial or replacement collaring event
mep_source_failure	Collar or GPS source failure report

9 columns are retained: id, time, event_type, event_category, reported_by, serial_number, geometry, event_details, priority_label (force_point_geometry: true, include_null_geometry: true). The event_details JSON column is normalised and the following fields are renamed:

Raw column	Renamed to
event_details__pic	pic
event_details__region	region
event_details__source	source
event_details__details	details
event_details__subject	groupby_col
event_details__source_id	source_id
event_details__subject_id	subject_name
event_details__collaring_type	collaring_type
event_details__collaring_reason	collaring_reason
event_details__collar_checked_by	collar_checked_by

Events are then split by subject_name to align with the per-subject processing pipeline. The required column for splitting is **event_details__subject_id**.

5. Trajectory Processing

5.1 Relocations to trajectories

Relocations are converted to trajectory segments via **relocations_to_trajectory** using a user-configurable **custom_trajectory_segment_filter**. The filter allows setting maximum time gap and speed thresholds to break trajectories at implausible movement gaps.

5.2 Temporal index

add_temporal_index assigns a temporal index to each trajectory segment using **segment_start** as the time column (cast_to_datetime: true, format: mixed). The index is used for time-based seasonal aggregation.

5.3 Speed classification

Segment speeds (speed_kmhr) are classified into 6 equal-interval bins via **apply_classification** (scheme: equal_interval, k: 6). Labels show ranges to 1 decimal place with a ' km/h' suffix. A 6-colour diverging ramp is then applied:

Bin rank	Hex color	Visual meaning
1 (slowest)	#1a9850	Dark green
2	#91cf60	Light green
3	#d9ef8b	Yellow-green
4	#fee08b	Yellow
5	#fc8d59	Orange
6 (fastest)	#d73027	Red

After classification, extra__ prefix columns are renamed (extra__hex_color→hex_color, extra__subject_name→subject_name, extra__subject_sex→subject_sex, extra__subject_subtype→subject_subtype, extra__created_at→created_at) and trajectories are split by subject_name for per-subject map rendering.

6. Seasonal Analysis

Seasonal windows are derived per subject using Google Earth Engine NDVI data. The workflow uses the subject's ETD polygon as the region of interest (ROI) for NDVI extraction.

6.1 ETD home range

calculate_elliptical_time_density computes an ETD raster for each subject from its split trajectory GeoDataFrame. Parameters:

Parameter	Value
Percentiles	50, 60, 70, 80, 90, 95, 99.9
CRS	ESRI:53042 (World Azimuthal Equidistant)
Cell size	Auto-scale
max_speed_factor	1.05
expansion_factor	1.3
nodata_value	NaN
band_count	1

The ETD is reprojected to EPSG:4326 for map rendering. A RdYlGn colormap is applied to the percentile column (etd_percentile_colors) and used to colour the GeoJSON polygon layer.

6.2 Seasonal window detection

determine_season_windows calls Google Earth Engine to identify wet and dry season boundaries for the analysis time range, using the ETD polygon extent as the ROI. Results are persisted as CSV per subject.

6.3 Season labelling

create_seasonal_labels assigns a season label (wet/dry) to each trajectory segment and relocation based on the GEE-derived windows. These labels drive the seasonal home range calculation and all seasonal overlay plots.

6.4 Seasonal home range

calculate_seasonal_home_range computes a 99.9th-percentile ETD polygon grouped by season (groupby_cols: ['season'], auto-scale). A 2-color scheme is applied:

Season index	Color	Visual
0 (first season)	#00bfff	Deep sky blue
1 (second season)	#ff7f50	Coral

A Minimum Convex Polygon (MCP) is generated from the trajectory GDF (**generate_mcp_gdf**, planar CRS ESRI:53042) and overlaid as a hot-pink outline (#ff1493, opacity 0.35, unfilled) on the seasonal home range map.

7. Maps

All three map types are rendered per subject via **mapvalues**. Each map overlays the three LandDx land-use layers (2.3). Map extent is fixed internally — **envelope_gdf** (expansion_factor: 1.25) pads the subject's data extent, and **compute_view_state_from_gdf** (pitch: 0, bearing: 0, max_zoom: 15) derives the view state; neither is user-configurable. Screenshots use **html_to_png** with wait_for_timeout: 10 ms, device_scale_factor: 2.0, max_concurrent_pages: 1, full_page: false.

7.1 Speed map

Built from trajectory segments as a **create_path_layer**. Segments are sorted ascending by speed_bins before rendering (slowest drawn first, fastest on top).

Parameter	Value
Layer type	Path (line segments)
Color column	speed_bins_colormap (6-bin RdYlGn)
Line width	2.25 px (min 2 px / max 8 px)
Opacity	0.45
Cap / joint style	Rounded
Screenshot size	1280 × 720 px
Output	_speedmap.html / .png

7.2 Home range map (ETD)

ETD percentile polygons are rendered as filled GeoJSON layers. All 7 percentile contours (50–99.9) are stacked with the outermost contour at the bottom.

Parameter	Value
Layer type	GeoJSON polygon
Color column	etd_percentile_colors (RdYlGn on percentile)
Opacity	0.45
Line width	1.55 px
Screenshot size	602 × 855 px
Output	_homerange.html / .png

7.3 Seasonal home range map

Two seasonal ETD polygons (99.9th percentile, wet vs. dry) are rendered as filled GeoJSON layers. The MCP polygon is overlaid as an unfilled hot-pink outline.

Layer	Color	Opacity	Notes
Seasonal HR (wet)	#00bfff	0.15	Deep sky blue fill, black outline
Seasonal HR (dry)	#ff7f50	0.15	Coral fill, black outline
MCP outline	#ff1493	0.35	Unfilled, hot-pink stroke only

Screenshot size: 602 × 855 px. Output: **<subject>_seasonal_home_range.html / .png**

8. Seasonal Plots

Four time-series plots are generated per subject. Each plot overlays the GEE-derived seasonal windows as background bands. All plot screenshots are rendered at **2238 × 450 px** with wait_for_timeout: 5 ms (fast static HTML).

Plot	Task	X axis	Y axis	Output file
Net Square Displacement	draw_season_nsd_plot	Time	NSD from first fix (km²)	_nsd_seasonal_plot.html/.png
Speed	draw_season_speed_plot	Time	Speed (km/h)	_speed_seasonal_plot.html/.png
Collar Events	draw_season_collared_plot	Time	Event type (collar check / collaring / source failure)	_collared_subject_plot.html/.png
MCP Asymptote	draw_season_mcp_plot	Cumulative fixes	MCP area (km²)	_mcp_asymptote_plot.html/.png

Note: The collar event plot uses filter_col: subject_name to match events from the MEP event pipeline (mep_collar_check, mep_collaring, mep_source_failure) to the current subject's relocations.

9. Statistics and Occupancy Analysis

9.1 Subject statistics

compute_subject_stats derives movement metrics per subject by combining the ETD GDF, trajectory GDF, and maturity-flagged subject metadata (grouped by subject_name). Results are persisted as CSV and rendered as a "Subject Stats" table widget:

Field	Unit	Description
subject_id	—	Subject identifier
mature	—	Whether the subject meets the 6-month maturity threshold
mcp	km²	Minimum Convex Polygon area
etd	km²	Elliptical Time Density 99.9th percentile area
time_tracked_days	days	Days between first and last fix
time_tracked_years	years	Years between first and last fix
distance_travelled	km	Total path length of trajectory
max_displacement	km	Maximum distance from first fix
night_day_ratio	—	Ratio of nocturnal to diurnal fixes

MCP, ETD, distance travelled, max displacement, and night/day ratio are also summed across the subject group (dataframe_column_sum) and exposed individually as single-value dashboard widgets.

9.2 Regional occupancy

compute_subject_occupancy intersects the subject's ETD polygon with LandDx region templates in planar CRS ESRI:53042 to compute the proportion of home range falling in each land category. LandDx polygons are first indexed via **build_ldx_template_region_lookup** and **compute_template_regions**.

Occupancy column	Unit	Description
national_pa_use	%	National protected area use (National Park + National Reserve)
community_pa_use	%	Community conservancy use
crop_raid_percent	%	Agricultural land use
kenya_use	%	Overall Kenya coverage
unprotected	%	Unprotected land use

All five occupancy columns are written to CSV per subject and rendered together as a single "Subject Occupancy" table widget on the dashboard (not as individual single-value widgets).

10. Word Mapbook

The final report is a merged Word document assembled from a cover page and one section page per subject.

10.1 Cover page

prepare_cover_metadata assembles the cover context, including the organisation logo downloaded from Dropbox (MEP-logo-dark-linear.png):

Template	cer_cover_page.docx (downloaded from Dropbox)
Output	subject_tracking_cover_page.docx
Fields	org_logo_path (downloaded MEP logo), report_period (time range), prepared_by='Ecoscope', time_generated_format='%Y-%m-%d %H:%M:%S'

10.2 Per-subject section

Template	mep_subject_template_two.docx (downloaded from Dropbox)
Output	.docx (one file per subject)

Each section is populated by **generate_subject_report**, which takes the per-subject split DataFrame directly (mapvalues over split_subject_by_group) and resolves the subject's photo, plots, maps, and CSV tables from **ECOSCOPE_WORKFLOWS_RESULTS** by filename convention, rather than being wired explicitly via zip_groupbykey.

10.3 Merge

merge_docx_documents merges the cover page (subject_tracking_cover_page.docx) with all per-subject section documents (order_by: name) into a single output file: **overall_report.docx**.

11. Dashboard Widgets

The final **gather_dashboard** call assembles 15 widget groups into an interactive Ecoscope dashboard. All widgets are per-subject grouped views merged via **merge_widget_views**.

11.1 Single-value widgets (5)

Widget title	Source column	Unit
MCP Area	mcp	km²
ETD	etd	km²
Distance Travelled	distance_travelled	km²
Max Displacement	max_displacement	km²
Night Day Ratio	night_day_ratio	—

11.2 Map and plot widgets (7)

Widget title	Source
Speed Map	persist_speedmap_html
Home Range	persist_homerange_html
Seasonal Home Range	persist_seasonal_home_range_html
Net Square Displacement (NSD)	persist_nsd_html_urls
Speed	persist_speed_html_urls
Collared Subject Plot	persist_collared_subject_plots
MCP Asymptote	persist_mcp_html_urls

11.3 Table widgets (3)

Widget title	Source
Subject Info	persist_subject_info_table_html
Subject Stats	persist_subject_stats_table_html
Subject Occupancy	persist_subject_occ_table_html

12. Output Files

All outputs are written to **$ECOSCOPE_WORKFLOWS_RESULTS**. Files marked with <*subject*> are produced once per subject in the group.

File	Description
_speedmap.html / .png	Speed map (1280×720 px screenshot)
_homerange.html / .png	ETD home range map (602×855 px screenshot)
_seasonal_home_range.html / .png	Seasonal home range + MCP overlay (602×855 px)
_nsd_seasonal_plot.html / .png	Net Square Displacement plot (2238×450 px)
_speed_seasonal_plot.html / .png	Speed seasonal plot (2238×450 px)
_collared_subject_plot.html / .png	Collar event timeline plot (2238×450 px)
_mcp_asymptote_plot.html / .png	MCP area asymptote plot (2238×450 px)
_subject_info.csv / _subject_info_table.html	Subject metadata (name, sex, DOB, status, region, notes, bio, distribution)
_subject_stats.csv / _subject_stats_table.html	Movement stats: maturity, days/years tracked, MCP, ETD, distance, max displacement, night/day ratio
_subject_occupancy.csv / _subject_occupancy_table.html	Occupancy percentages by land category
_seasonal_windows.csv	GEE-derived wet/dry season date windows
_profile.png	Subject profile photo downloaded from EarthRanger
.docx	Per-subject populated Word section
relocations.parquet / subjects.parquet	All subjects' cleaned relocations / metadata with maturity flag
events.parquet / trajectories.parquet	All fetched MEP collar events / classified trajectory segments
subject_tracking_cover_page.docx	Cover page (organisation logo, report period, prepared by)
overall_report.docx	Final merged Word mapbook (cover + all subject sections)

13. Workflow Execution Logic

13.1 Global skip conditions

The top-level **task-instance-defaults** block applies two skip conditions to every task unless overridden:

- **any_is_empty_df** — skips the task if any upstream DataFrame dependency is empty
- **any_dependency_skipped** — skips the task if any upstream task was skipped

A special override **skipif: conditions: [never]** is applied to widget creation tasks to ensure dashboard entries are always created even if the underlying data is empty.

13.2 mapvalues fan-out

The workflow uses **mapvalues** extensively to run per-subject processing in parallel. The fan-out starts at **split_traj_by_group**, **split_subject_by_group**, and **split_events_by_group**, which partition data by subject_name. Downstream tasks receive one data slice per subject and run independently. Key fan-out chains:

- split_traj_by_group → sort_trajs_by_speed → apply_speed_colormap → filter_speed_cols → generate_speedmap_layers → draw_speedmap
- split_traj_by_group → generate_etd → determine_seasonal_windows (GEE) → add_season_labels → seasonal_home_range
- split_subject_by_group → download_profile_pic, download_subject_info, subject_track_doc
- zip_traj_etd_gdf → generate_subject_stats
- zip_etd_subject_df → process_subject_occupancy

13.3 zip_groupbykey

groupbykey is used to pair multiple per-subject sequences by their shared key (subject_name) before passing to mapvalues tasks. Examples:

- zip_etd_with_traj — pairs seasonal windows with trajectory slices for season labelling
- zip_relocs_with_seasons — pairs seasonal windows with relocation slices for NSD/speed/collared/MCP plots
- zip_speedmap_with_viewstate, zip_hr_with_viewstate, zip_seasonal_hr_with_viewstate — pair combined map layers with view state for draw_map
- zip_traj_etd_gdf, zip_etd_subject_df — pair trajectory/ETD/subject slices for stats and occupancy calculations

13.4 Screenshot timing

All map and plot screenshots use a uniform **html_to_png** configuration: wait_for_timeout: 10 ms, device_scale_factor: 2.0, max_concurrent_pages: 1, full_page: false.

Output	Width × Height
Speed map	1280 × 720 px
Home range map	602 × 855 px
Seasonal home range map	602 × 855 px
NSD / speed / collared / MCP plots	2238 × 450 px

14. Software Versions

Package	Version pinned
ecoscope-platform	>=2.15.0, <2.16.0
ecoscope-workflows-ext-custom	0.1.0rc14.*
ecoscope-workflows-ext-ste	0.0.0rc1.*
ecoscope-workflows-ext-mep	1.0.0.*
pydeck	0.9.2
opentelemetry-sdk	>=1.20.0, <2.0.0

Note: All packages are resolved from the prefix.dev Ecoscope conda channels. The wildcard patch-version pin (.*) allows bug-fix releases to be picked up automatically while keeping minor and major versions locked.